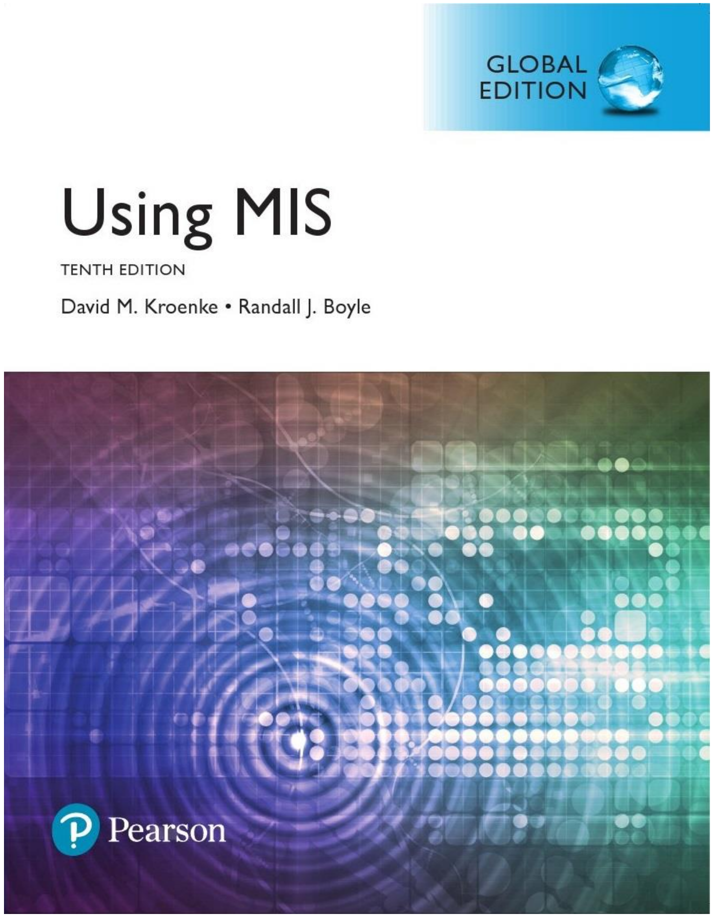
HỆ THỐNG THÔNG TIN QUẢN LÝ



Chương 9

Các hệ thống Kinh doanh thông minh

"Data Analysis, Where You Don't Know the Second Question to Ask Until You See the Answer to the First One."

- Having great success with employers interested in tracking exercise data.
- Wants to match users to personal trainers in same locale.
- Earn referral fee.
- How to track them? Mailing address? IP address?
- Got data and Excel to start.
- Serious data mining needs a data mart.

Study Questions

Q9-1 How do organizations use business intelligence (BI) systems?

Q9-2 What are the three primary activities in the BI process?

Q9-3 How do organizations use data warehouses and data marts to acquire data?

Q9-4 How do organizations use reporting applications?

Q9-5 How do organizations use data mining applications?

Q9-6 How do organizations use Big Data applications?

Q9-7 What is the role of knowledge management systems?

Q9-8 What are the alternatives for publishing BI?

Q9-9 2027?

Components of Business Intelligence (BI) Systems

Q9-1 How do organizations use business intelligence (BI) systems?

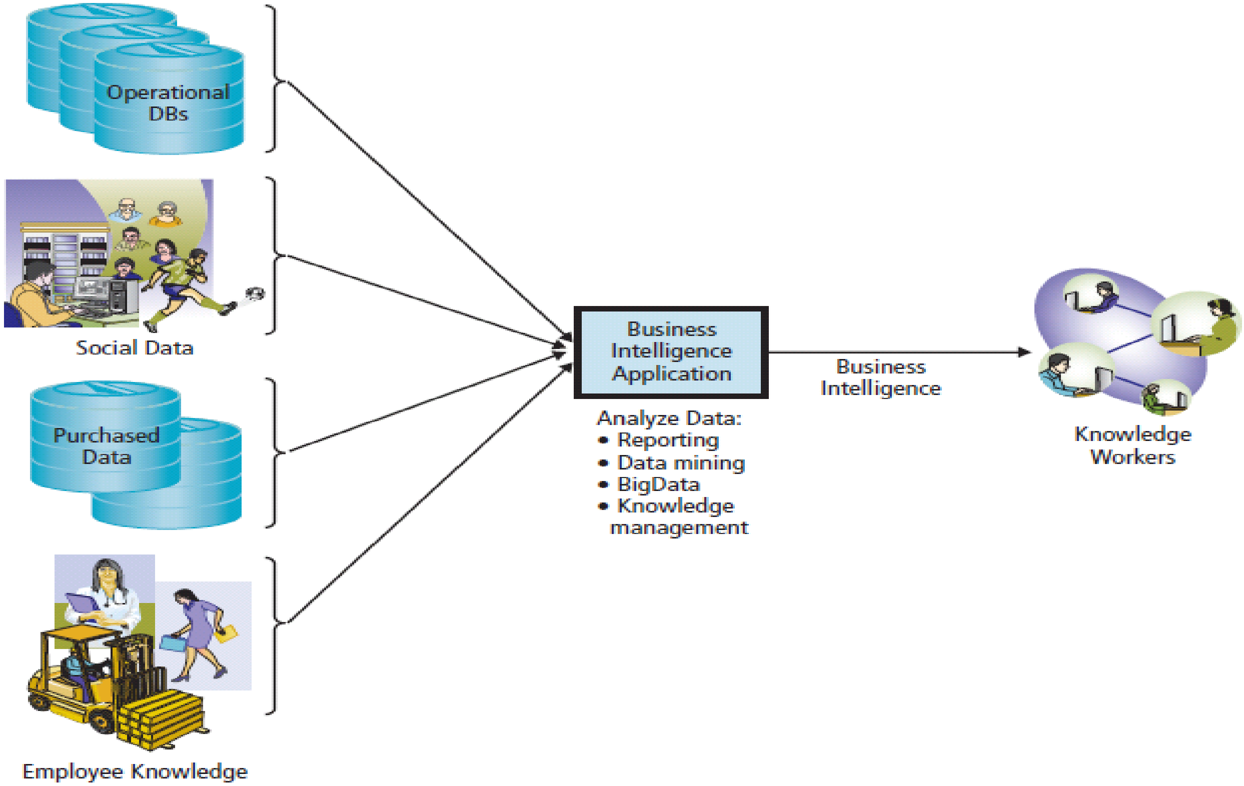


Figure 9-1 Components of a Business Intelligence System

How Do Organizations Use BI?

Q9-1 How do organizations use business intelligence (BI) systems?

Task	ARES Example	Falcon Security Example
Project Management	Create partnership programs between ARES users and local health clubs.	Expand geographically.
Problem Solving	How can we increase revenue from health clubs?	How can we save money by rerouting drone flights?
Deciding	Which health club is closest to each user? Refer users to local trainers.	Which drones and related equipment are in need of maintenance?
Informing	In what ways are clients using the new system?	How do sales compare to our sales forecast?

Figure 9-2 Example Uses of Business Intelligence

What Are Typical Uses for BI?

- Identifying changes in purchasing patterns
  - Important life events change what customers buy.
- Entertainment
  - Netflix has data on watching, listening, and rental habits.
  - Classify customers by viewing patterns.
- **Predictive policing**
  - Analyze data on past crimes - location, date, time, day of week, type of crime, and related data.

Just-in-Time Medical Reporting

- Example of real time data mining and reporting.
- Injection notification services
  - Software analyzes patient's records; if injections needed, recommends as exam progresses.
- Blurry edge of medical ethics.

Three Primary Activities in the BI Process

Q9-2 What are the three primary activities in the BI process?

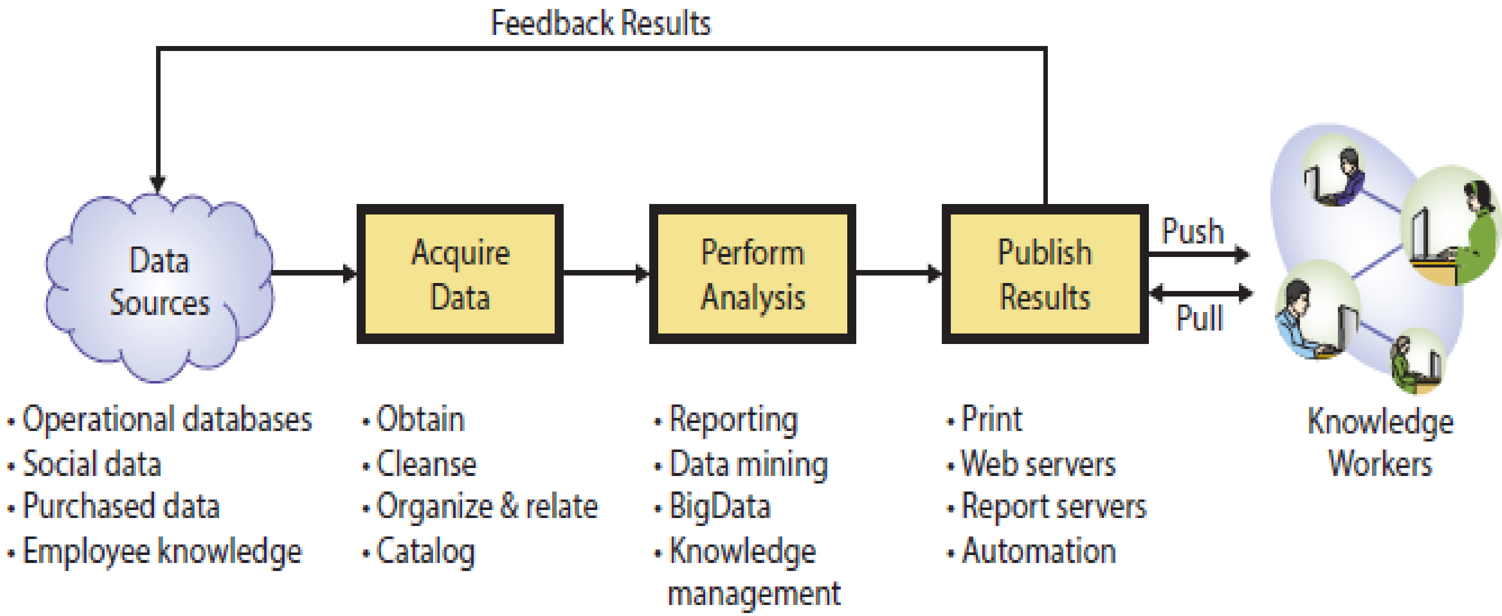


Figure 9-3 Three Primary Activities in the BI Process

Using Business Intelligence to Find Candidate Parts at Falcon Security

- Identify parts that might qualify.
  - Provided by vendors who make part design files available for sale.
  - Purchased by larger customers.
  - Frequently ordered parts.
  - Ordered in small quantities.
- Used part weight and price surrogates for simplicity.

Acquire Data: Extracted Order Data

Q9-2 What are the three primary activities in the BI process?

- Query

  Sales (CustomerName, Contact, Title, Bill Year, Number Orders, Units, Revenue, Source, PartNumber)

  Part (PartNumber, Shipping Weight, Vendor)

Island Biking	John Steel	Marketing Manager	2017	10	39	$195.22	AWS
Island Biking	John Steel	Marketing Manager	2016	14	59	$438.81	Internet
Island Biking	John Steel	Marketing Manager	2016	21	55	$255.96	AWS
Island Biking	John Steel	Marketing Manager	2017	4	11	$85.55	Internet
Kona Riders	Renate Messne	Sales Representative	2014	43	54	$349.27	Internet
Kona Riders	Renate Messne	Sales Representative	2015	30	53	$362.45	Internet
Kona Riders	Renate Messne	Sales Representative	2016	1	2	$14.34	Internet
Lone Pine Crafters	Jaime Yorres	Owner	2017	4	14	$108.89	Internet
Lone Pine Crafters	Jaime Yorres	Owner	2017	2	2	$15.56	Internet
Lone Pine Crafters	Jaime Yorres	Owner	2018	2	2	$15.56	Internet
Moab Mauraders	Carlos Gonzále	Accounting Manager	2017	2	4	$4,106.69	Internet
Moab Mauraders	Carlos Gonzále	Accounting Manager	2017	3	7	$7,404.18	Internet
Moab Mauraders	Carlos Gonzále	Accounting Manager	2017	2	6	$6,346.44	Internet
Sedona Mountain Trails	Felipe Izquierd	Owner	2017	6	7	$73.46	Internet

Figure 9-4a Sample Extracted Data: Order Extract Table
Source: Microsoft Corporation

Sample Extracted Data: Part Data Table

Q9-2 What are the three primary activities in the BI process?

Part Data

ID	PartNumber	Shipping Weight	Vendor	Click to Add
9	200-219	7.28	DePARTures, Inc.	
22	200-225	3.61	DePARTures, Inc.	
23	200-227	5.14	DePARTures, Inc.	
11	200-207	9.23	DePARTures, Inc.	
28	200-205	4.11	DePARTures, Inc.	
29	200-211	4.57	DePARTures, Inc.	
10	200-213	1.09	DePARTures, Inc.	
37	200-223	3.61	DePARTures, Inc.	
45	200-217	1.98	DePARTures, Inc.	
2	200-209	10.41	DePARTures, Inc.	
3	200-215	1.55	DePARTures, Inc.	
47	200-221	10.85	DePARTures, Inc.	
42	200-203	3.20	DePARTures, Inc.	
17	300-1007	2.77	Desert Gear Supply	

Figure 9-4b Sample Extracted Data: Part Data Table
Source: Microsoft Corporation

Analyze Data

Q9-2 What are the three primary activities in the BI process?

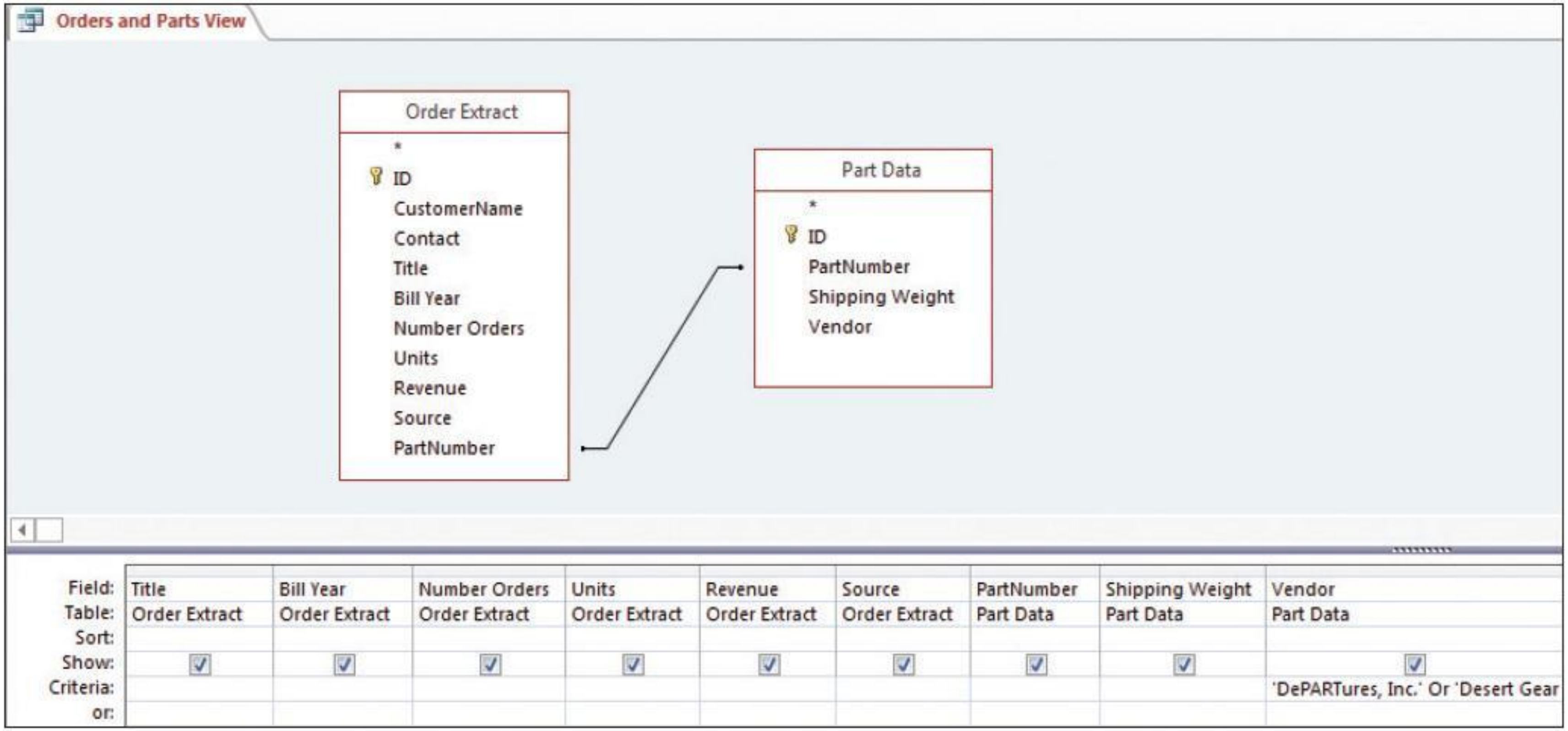


Figure 9-5 Joining Orders Extract and Filtered *Parts* Tables
Source: Microsoft Corporation

Sample Orders and Parts View Data

Q9-2 What are the three primary activities in the BI process?

Orders and Parts View

CustomerName	Contact	Title	Bill Year	Number Orders	Units	Revenue	Source	PartNumber	Shipping Weight	Vendor
Gordos Dirt Bikes	Sergio Gutiérrez	Sales Repres	2011	43	107	$26,234.12	Internet	100-108	3.32	Riley Manufacturing
Island Biking			2012	59	135	$25,890.62	Phone	500-2035	9.66	ExtremeGear
Big Bikes			2010	29	77	$25,696.00	AWS	700-1680	6.06	HyperTech Manufacturing
Lazy B Bikes			2009	19	30	$25,576.50	Internet	700-2280	2.70	HyperTech Manufacturing
Lone Pine Crafters	Carlos Hernández	Sales Repres	2012	1	0	$25,171.56	Internet	500-2030	4.71	ExtremeGear
Seven Lakes Riding	Peter Franken	Marketing Ma	2009	15	50	$25,075.00	Internet	500-2020	10.07	ExtremeGear
Big Bikes			2012	10	40	$24,888.00	Internet	500-2025	10.49	ExtremeGear
B' Bikes	Georg Pipps	Sales Manage	2012	14	23	$24,328.02	Internet	700-1680	6.06	HyperTech Manufacturing
Eastern Connection	Isabel de Castro	Sales Repres	2012	48	173	$24,296.17	AWS	100-105	10.73	Riley Manufacturing
Big Bikes	Carine Schmitt	Marketing Ma	2009	22	71	$23,877.48	AWS	500-2035	9.66	ExtremeGear
Island Biking	Manuel Pereira	Owner	2011	26	45	$23,588.86	Internet	500-2045	3.22	ExtremeGear
Mississippi Delta Riding	Rene Phillips	Sales Repres	2012	9	33	$23,550.25	Internet	700-2180	4.45	HyperTech Manufacturing
Uncle's Upgrades			2012	9	21	$22,212.54	Internet	700-1680	6.06	HyperTech Manufacturing
Big Bikes			2010	73	80	$22,063.92	Phone	700-1680	6.06	HyperTech Manufacturing
Island Biking			2012	18	59	$22,025.88	Internet	100-108	3.32	Riley Manufacturing
Uncle's Upgrades			2011	16	38	$21,802.50	Internet	500-2035	9.66	ExtremeGear

Figure 9-6 Sample Orders and Parts View Data
Source: Microsoft Corporation

Creating Customer Summary Query

Q9-2 What are the three primary activities in the BI process?

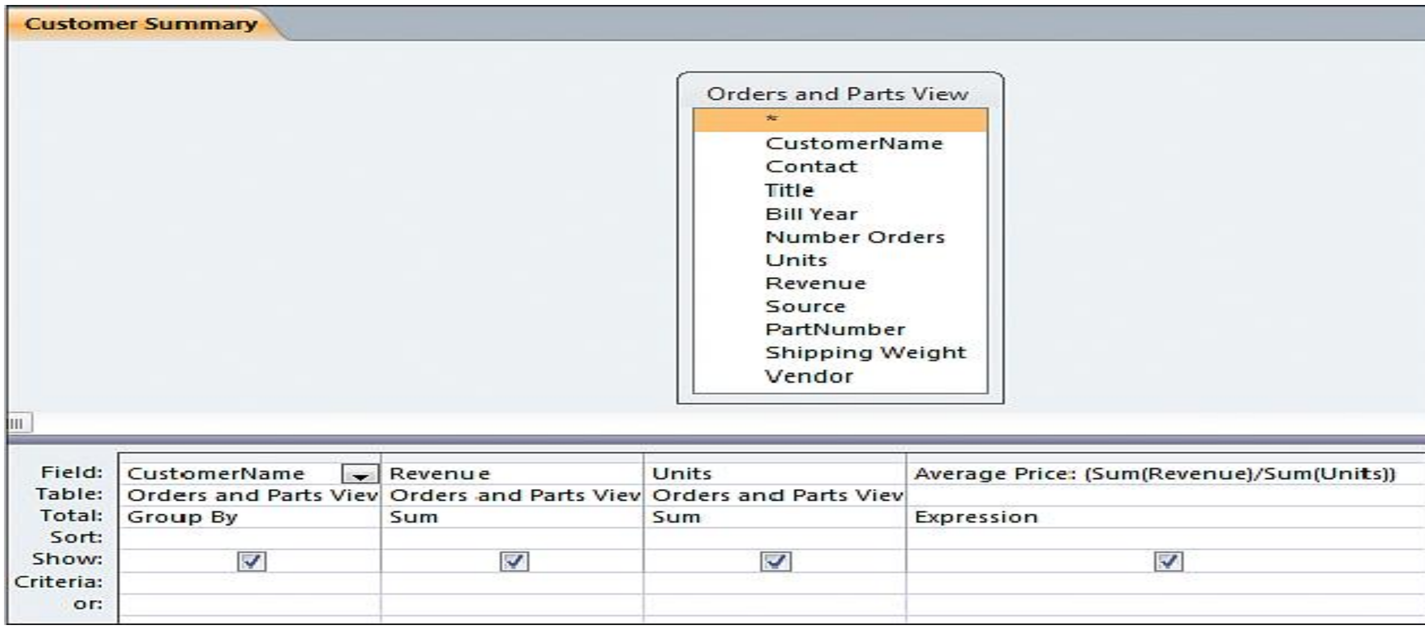


Figure 9-7 Creating the Customer Summary Query
Source: Microsoft Corporation

Customer Summary

Q9-2 What are the three primary activities in the BI process?

Customer Summary

CustomerName	SumOfRevenue	SumOfUnits	Average Price
Great Lakes Machines	$1,760.47	142	12.3976535211268
Seven Lakes Riding	$288,570.71	5848	49.3451963919289
Around the Horn	$16,669.48	273	61.0603611721612
Dewey Riding	$36,467.90	424	86.0092018867925
Moab Mauraders	$143,409.27	1344	106.7033234375
Gordos Dirt Bikes	$113,526.88	653	173.854335068913
Mountain Traders	$687,710.99	3332	206.395855432173
Hungry Rider Off-road	$108,602.32	492	220.736416056911
Eastern Connection	$275,092.28	1241	221.669848186946
Mississippi Delta Riding	$469,932.11	1898	247.593315542676
Island Biking	$612,072.64	2341	261.457770098249

Figure 9-8 Customer Summary
Source: Microsoft Corporation

Qualifying Parts Query Design

Q9-2 What are the three primary activities in the BI process?

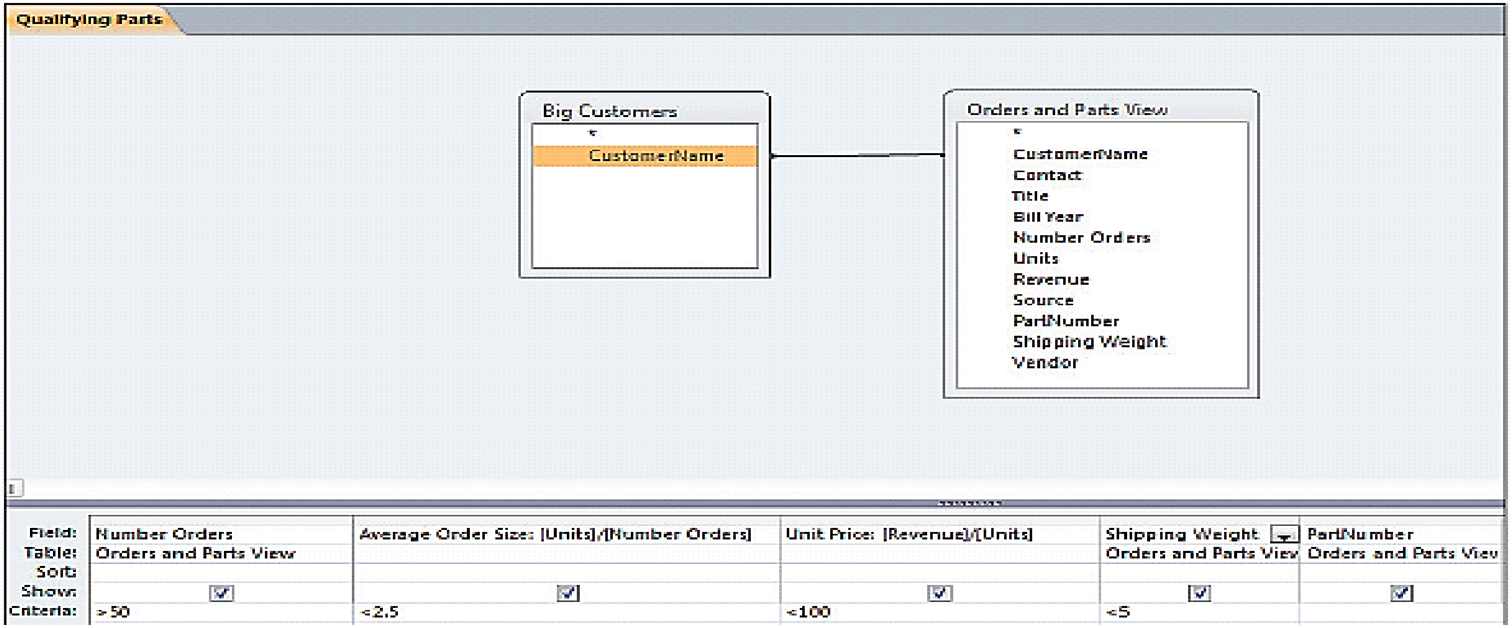


Figure 9-9 Qualifying Parts Query Design
Source: Microsoft Corporation

Publish Results: Qualifying Parts Query Results

Q9-2 What are the three primary activities in the BI process?

Qualifying Parts

Number Orders	Average Order Size	Unit Price	Shipping Weight	PartNumber
275	1	9.14173854545455	4.14	300-1016
258	1.87596899224806	7.41284524793388	4.14	300-1016
110	1.18181818181818	6.46796923076923	4.11	200-205
176	1.66477272727273	12.5887211604096	4.14	300-1016
139	1.0431654676259	6.28248965517241	1.98	200-217
56	1.83928571428571	6.71141553398058	1.98	200-217
99	1.02020202020202	7.7775	3.20	200-203
76	2.17105263157895	12.0252206060606	2.66	300-1013

Figure 9-10 Qualifying Parts Query Results
Source: Microsoft Corporation

Publish Results: Sales History for Selected Parts

Q9-2 What are the three primary activities in the BI process?

Revenue Potential

Total Orders	Total Revenue	PartNumber
3987	$84,672.73	300-1016
2158	$30,912.19	200-211
1074	$23,773.53	200-217
548	$7,271.31	300-1007
375	$5,051.62	200-203
111	$3,160.86	300-1013
139	$1,204.50	200-205

Figure 9-11 Sales History for Selected Parts
Source: Microsoft Corporation

MIS-Diagnosis

Ethics Guide

- Doctors are relying more and more on artificial intelligence (AI)-driven expert systems to select the most appropriate medications and treatments.
- Ordered to improve the system's "perception" of the company's drugs.
- Minor modifications to the drug's profile made a big difference.
  - But some of the numbers he used to modify the profile were *not* accurate.
  - The changes would warrant a *regulatory review*.

MIS-Diagnosis (cont'd)

Ethics Guide

- Suppose the company alters the drug profile.
- Would the company be liable if something happened to a patient who took the drug based on *altered* information?
- Do you think that manipulating the recommendation of an AI system even though the new recommendation may be for the better drug is ethical according to the categorical imperative, and utilitarian perspective?

Using Data Warehouses and Data Marts to Acquire Data

- Functions of a data warehouse
  - Obtain data from operational, internal and external databases.
  - Cleanse data.
  - Organize and relate data.
  - Catalog data using metadata.

Components of a Data Warehouse

Q9-3 How do organizations use data warehouses and data marts to acquire data?

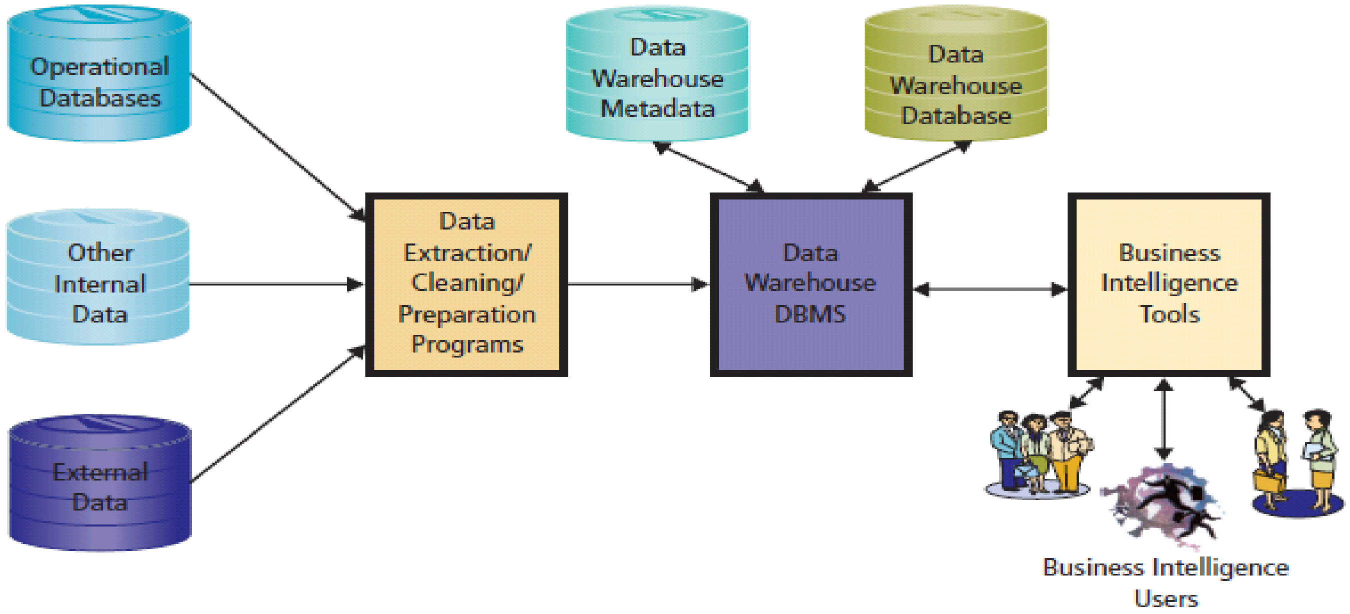


Figure 9-12 Components of a Data Warehouse

Examples of Consumer Data That Can Be Purchased

Q9-3 How do organizations use data warehouses and data marts to acquire data?

- Name, address, phone
- Age
- Gender
- Ethnicity
- Religion
- Income
- Education
- Voter registration
- Home ownership
- Vehicles
- Magazine subscriptions
- Hobbies
- Catalog orders
- Marital status, life stage
- Height, weight, hair and eye color
- Spouse name, birth date
- Children's names and birth dates

Figure 9-13 Examples of Consumer Data That Can Be Purchased

Possible Problems with Source Data

Q9-3 How do organizations use data warehouses and data marts to acquire data?

- Dirty data
- Missing values
- Inconsistent data
- Data not integrated
- Wrong granularity
  - Too fine
  - Not fine enough
- Too much data
  - Too many attributes
  - Too many data points

Figure 9-14 Possible Problems with Source Data

Data Warehouses Versus Data Marts

Q9-3 How do organizations use data warehouses and data marts to acquire data?

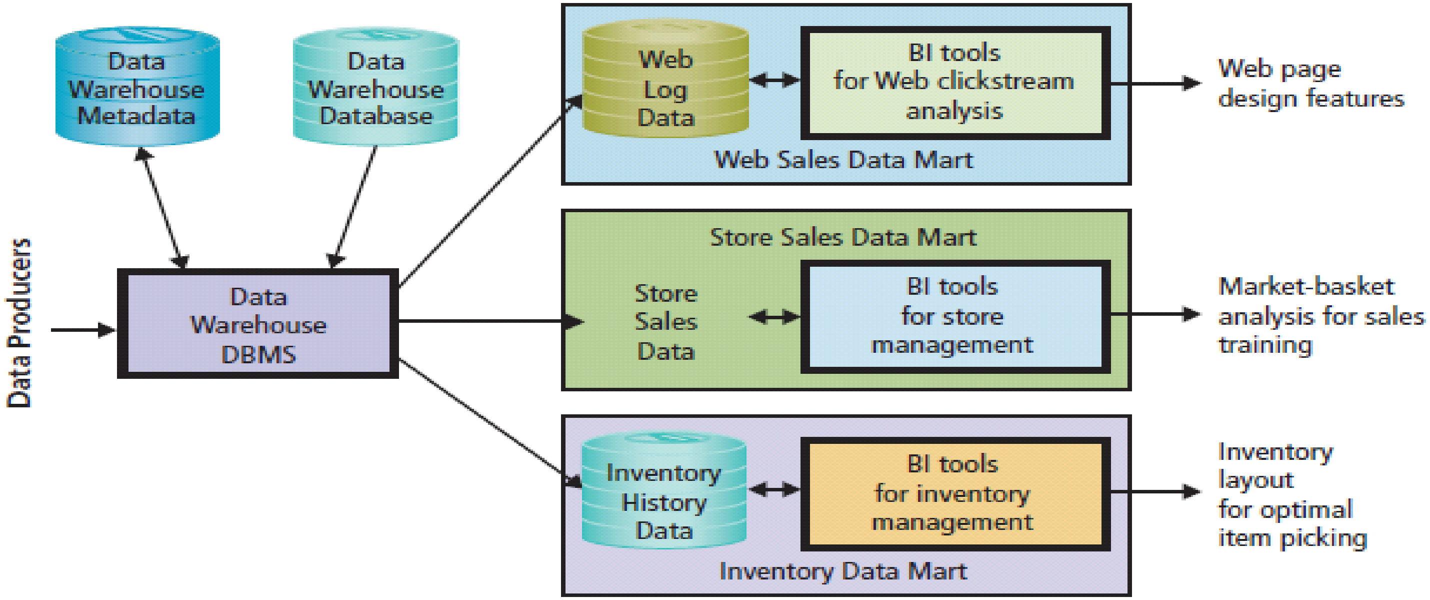


Figure 9-15 Data Mart Examples

Reporting Applications

Q9-4 How do organizations use reporting applications?

- Create meaningful information from disparate data sources.
- Deliver information to user on time.
- Basic operations:
  1. Sorting
  2. Filtering
  3. Grouping
  4. Calculating
  5. Formatting

RFM Analysis: Example RFM Scores

Q9-4 How do organizations use reporting applications?

- How *recently* (R) a customer has ordered
- How *frequently* (F) a customer ordered
- How much *money* (M) the customer has spent

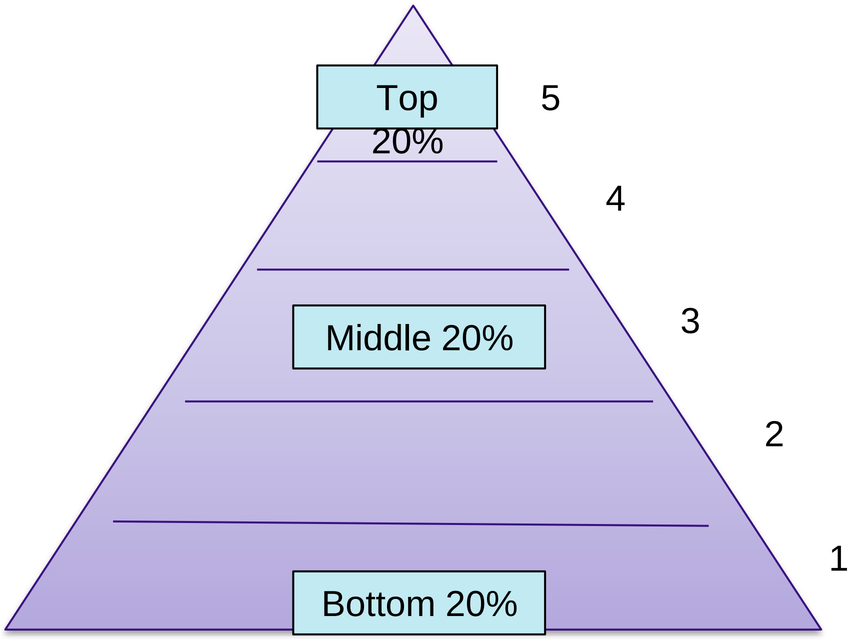
Customer	RFM Score
Big 7 Sports	1 1 3
St. Louis Soccer Club	5 1 1
Miami Municipal	5 4 5
Central Colorado State	3 3 3

Figure 9-16 Example RFM Scores

RFM Analysis Classification Scheme

Q9-4 How do organizations use reporting applications?

- To produce an RFM score:
  - Sort customer purchase records by date of most recent (R) purchase.
  - Divide sorts into quintiles.
  - Give customers a score of 1 to 5.
- Process is repeated for Frequently and Money.



Example of Grocery Sales OLAP Report

- OLAP Product Family by Store Type
  - http://www.tableausoftware.com

	A	B	C	D	E	F	G
1							
2							
3	Store Sales Net	Store Type					
4	Product Family	Deluxe Supermarket	Gourmet Supermarket	Mid-Size Grocery	Small Grocery	Supermarket	Grand Total
5	Drink	$8,119.05	$2,392.83	$1,409.50	$685.89	$16,751.71	$29,358.98
6	Food	$70,276.11	$20,026.18	$10,392.19	$6,109.72	$138,960.67	$245,764.87
7	Non-Consumable	$18,884.24	$5,064.79	$2,813.73	$1,534.90	$36,189.40	$64,487.05
8	Grand Total	$97,279.40	$27,483.80	$14,615.42	$8,330.51	$191,901.77	$339,610.90

Figure 9-17 Example Grocery Sales OLAP Report
Source: Microsoft Corporation

Example of Expanded Grocery Sales OLAP Report

Q9-4 How do organizations use reporting applications?

- Drilling down

1									
2									
3	Store Sales Net			Store Type					
4	Product Family	Store Country	Store State	Deluxe Superma	Gourmet Supermar	Mid-Size Groce	Small Grocery	Supermarket	Grand Total
5	Drink	USA	CA		$2,392.83		$227.38	$5,920.76	$8,540.97
6			OR	$4,438.49				$2,862.45	$7,300.94
7			WA	$3,680.56		$1,409.50	$458.51	$7,968.50	$13,517.07
8		USA Total		$8,119.05	$2,392.83	$1,409.50	$685.89	$16,751.71	$29,358.98
9	Drink Total			$8,119.05	$2,392.83	$1,409.50	$685.89	$16,751.71	$29,358.98
10	Food	USA	CA		$20,026.18		$1,960.53	$47,226.11	$69,212.82
11			OR	$37,778.35				$23,818.87	$61,597.22
12			WA	$32,497.76		$10,392.19	$4,149.19	$67,915.69	$114,954.83
13		USA Total		$70,276.11	$20,026.18	$10,392.19	$6,109.72	$138,960.67	$245,764.87
14	Food Total			$70,276.11	$20,026.18	$10,392.19	$6,109.72	$138,960.67	$245,764.87
15	Non-Consumable	USA	CA		$5,064.79		$474.35	$12,344.49	$17,883.63
16			OR	$10,177.89				$6,428.53	$16,606.41
17			WA	$8,706.36		$2,813.73	$1,060.54	$17,416.38	$29,997.01
18		USA Total		$18,884.24	$5,064.79	$2,813.73	$1,534.90	$36,189.40	$64,487.05
19	Non-Consumable Total			$18,884.24	$5,064.79	$2,813.73	$1,534.90	$36,189.40	$64,487.05
20	Grand Total			$97,279.40	$27,483.80	$14,615.42	$8,330.51	$191,901.77	$339,610.90

Figure 9-18 Example of Expanded Grocery Sales OLAP Report
Source: Microsoft Corporation

Example of Drilling Down into Expanded Grocery Sales OLAP Report

Q9-4 How do organizations use reporting applications?

1										
2										
3	Store Sales Net				Store Type					
4	Store Country	Store Sta	Store City	Product Family	Deluxe Super	Gourmet Supermar	Mid-Size Groce	Small Grocery	Supermarket	Grand Total
5	USA	CA	Beverly Hills	Drink		$2,392.83				$2,392.83
6				Food		$20,026.18				$20,026.18
7				Non-Consumable		$5,064.79				$5,064.79
8			Beverly Hills Total			$27,483.80				$27,483.80
9			Los Angeles	Drink					$2,870.33	$2,870.33
10				Food					$23,598.28	$23,598.28
11				Non-Consumable					$6,305.14	$6,305.14
12			Los Angeles Total						$32,773.74	$32,773.74
13			San Diego	Drink					$3,050.43	$3,050.43
14				Food					$23,627.83	$23,627.83
15				Non-Consumable					$6,039.34	$6,039.34
16			San Diego Total						$32,717.61	$32,717.61
17			San Francisco	Drink				$227.38		$227.38
18				Food				$1,960.53		$1,960.53
19				Non-Consumable				$474.35		$474.35
20			San Francisco Total					$2,662.26		$2,662.26
21		CA Total				$27,483.80		$2,662.26	$65,491.35	$95,637.41
22		OR		Drink	$4,438.49				$2,862.45	$7,300.94
23				Food	$37,778.35				$23,818.87	$61,597.22
24				Non-Consumable	$10,177.89				$6,428.53	$16,606.41
25		OR Total			$52,394.72				$33,109.85	$85,504.57
26		WA		Drink	$3,680.56		$1,409.50	$458.51	$7,968.50	$13,517.07
27				Food	$32,497.76		$10,392.19	$4,149.19	$67,915.69	$114,954.83
28				Non-Consumable	$8,706.36		$2,813.73	$1,060.54	$17,416.38	$29,997.01
29		WA Total			$44,884.68		$14,615.42	$5,668.24	$93,300.57	$158,468.91
30	USA Total				$97,279.40	$27,483.80	$14,615.42	$8,330.51	$191,901.77	$339,610.90
31	Grand Total				$97,279.40	$27,483.80	$14,615.42	$8,330.51	$191,901.77	$339,610.90

Figure 9-19 Example of Drilling Down into Expanded Grocery Sales OLAP Report
Source: Microsoft Corporation

Convergence of Disciplines

Q9-5 How do organizations use data mining applications?

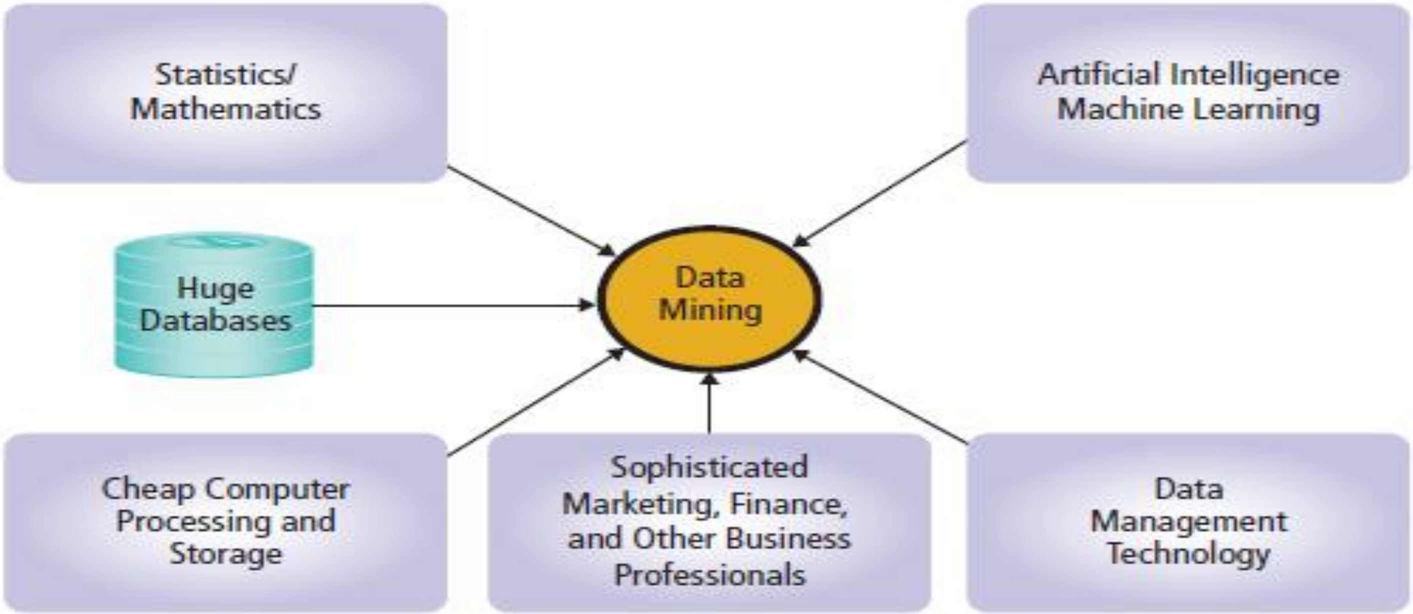


Figure 9-20 Source Disciplines of Data Mining

Unsupervised Data Mining

- No a priori hypothesis or model.
- Findings obtained solely by data analysis.
- Hypothesized model created to explain patterns found.
- Example: Cluster analysis.

Supervised Data Mining

Q9-5 How do organizations use data mining applications?

- Uses a priori model.
- Prediction, such as regression analysis.
- Ex: CellPhoneWeekendMinutes

= (12 + (17.5*CustomerAge)+(23.7*NumberMonthsOfAccount)

= 12 + 17.5*21 + 23.7*6 = **521.7 minutes**

Market-Basket Analysis

- Market-basket analysis
  - Identify sales patterns in large volumes of data.
  - Identify what products customers tend to buy together.
  - Computes probabilities of purchases.
  - Identify cross-selling opportunities.
    - Customers who bought fins also bought a mask.

Market-Basket Example: Dive Shop Transactions = 400

Q9-5 How do organizations use data mining applications?

	Mask	Tank	Fins	Weights	Dive Computer
Mask	270	10	250	10	90
Tank	10	200	40	130	30
Fins	250	40	280	20	20
Weights	10	130	[illegible]	130	10
Dive Computer	90	30	20	10	120
	Support				
Num Trans	[illegible]				
Mask	0.675	0.025	0.625	0.025	0.225
Tank	[illegible]	0.5	[illegible]	0.325	0.075
Fins	0.625	0.1	0.7	0.05	0.05
Weights	0.025	0.325	[illegible]	0.325	0.025
Dive Computer	0.225	0.075	0.05	0.025	0.3
	Confidence				
Mask	1	0.05	0.892857143	0.076923077	0.75
Tank	0.037037037	1	0.142857143	1	0.25
Fins	0.925925926	0.2	1	0.153846154	0.166666667
Weights	0.037037037	0.65	0.071428571	1	0.083333333
Dive Computer	0.333333333	0.15	0.071428571	0.076923077	1
	Lift (Improvement)				
Mask		0.074074074	1.322751323	0.113960114	1.111111111
Tank	0.074074074		0.285714286	2	0.5
Fins	1.322751323	0.285714286		0.21978022	0.238095238
Weights	0.113960114	2	0.21978022		0.256410256
Dive Computer	1.111111111	0.5	0.238095238	0.256410256	

Figure 9-21 Market-Basket Analysis at a Dive Shop
Source: Microsoft Corporation

Decision Trees

- Unsupervised data mining technique.
- Hierarchical arrangement of criteria to predict a value or classification.
- Basic idea
  - Select attributes most useful for classifying “pure groups.”
  - Creates decision rules.

Credit Score Decision Tree

Q9-5 How do organizations use data mining applications?

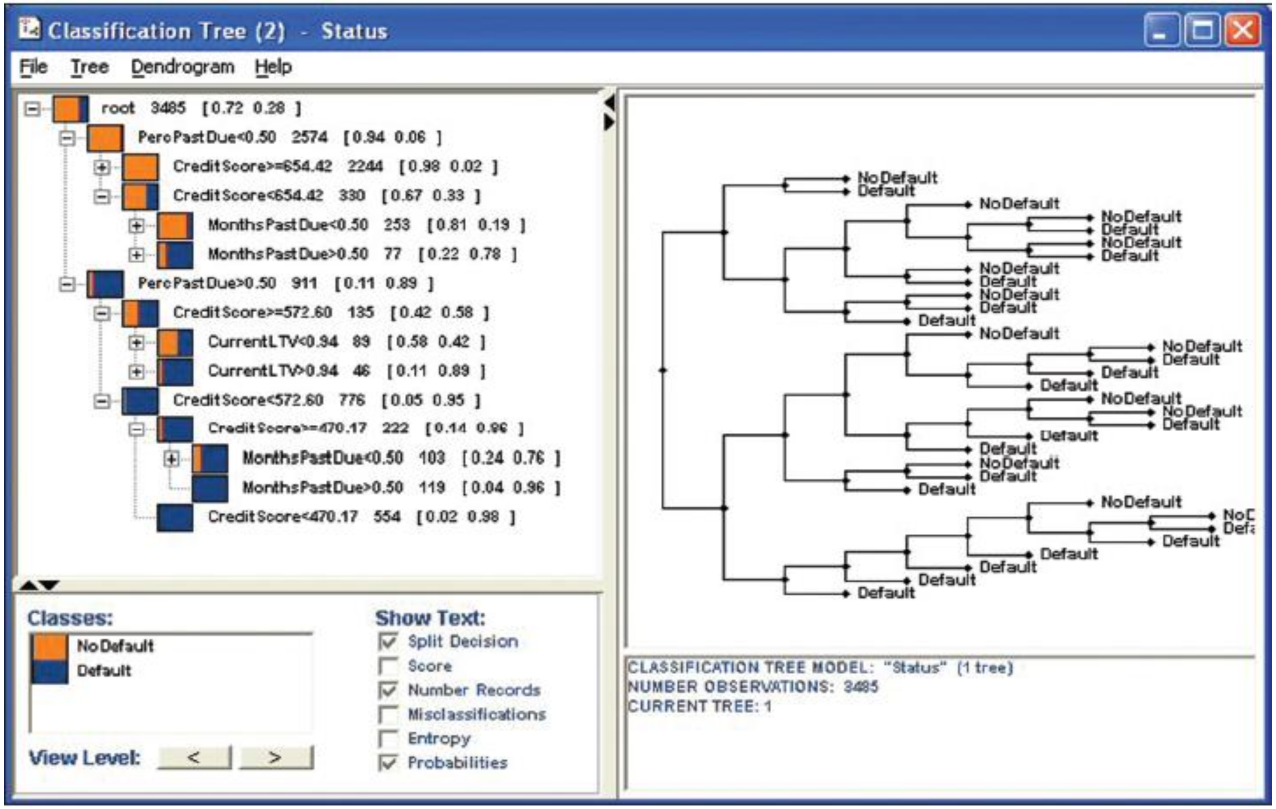


Figure 9-22 Credit Score Decision Tree

Decision Rules for Accepting or Rejecting Offer to Purchase Loans

- If percent past due is less than 50 percent, then accept loan.
    - If percent past due is greater than 50 percent *and*
    - If *CreditScore* is greater than 572.6 *and*
    - If *CurrentLTV* is less than .94, then accept loan.
- Otherwise, reject loan.

BI for Securities Trading?

So What?

- Quantitative applications using Big Data and BI.
  - Analyze immense amounts of data over a broad spectrum of sources.
  - Build and evaluate investment strategies.
- Two Sigma (*www.twosigma.com*)
  - Analyzes financial statements, developing news, Twitter activity, weather reports, other sources.
  - Develops and tests investment strategies.

Two Sigma's Five-Step Process

1. Acquire data
2. Create models
3. Evaluate models
4. Analyze risks
5. Place trades

- Does it work? Two Sigma and other firms claim it does.

Using Big Data Applications

Q9-6 How do organizations use Big Data applications?

- Huge volume – petabyte and larger.
- Rapid velocity – generated rapidly.
- Great variety
  - Structured data, free-form text, log files, graphics, audio, and video.

MapReduce Processing Summary

Q9-6 How do organizations use Big Data applications?

- Map Phase: Google search log broken into thousands of pieces

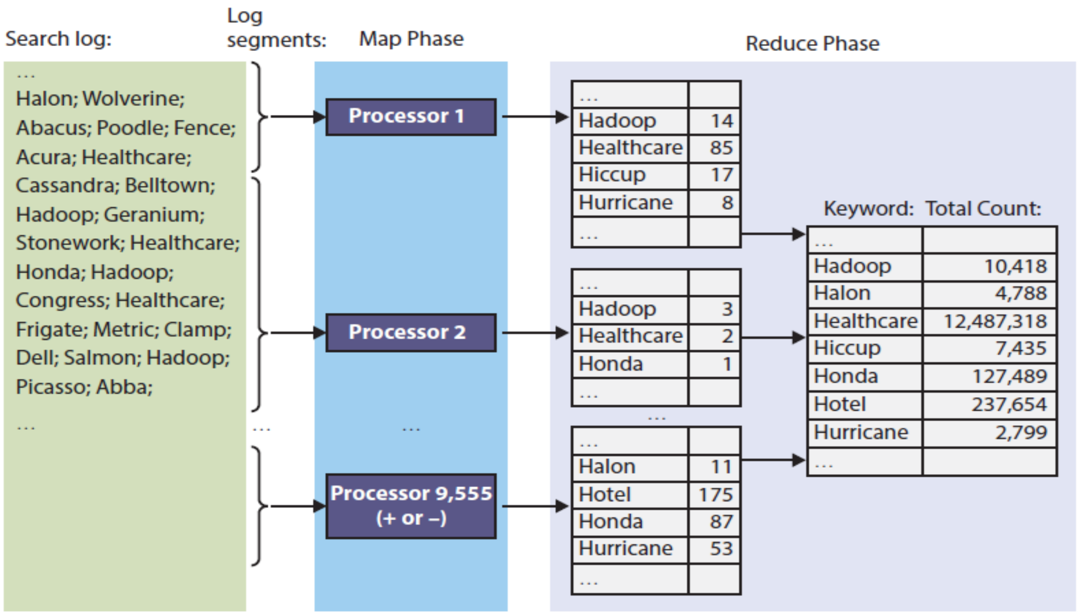


Figure 9-23 MapReduce Processing Summary

Google Trends on the Term Web 2.0

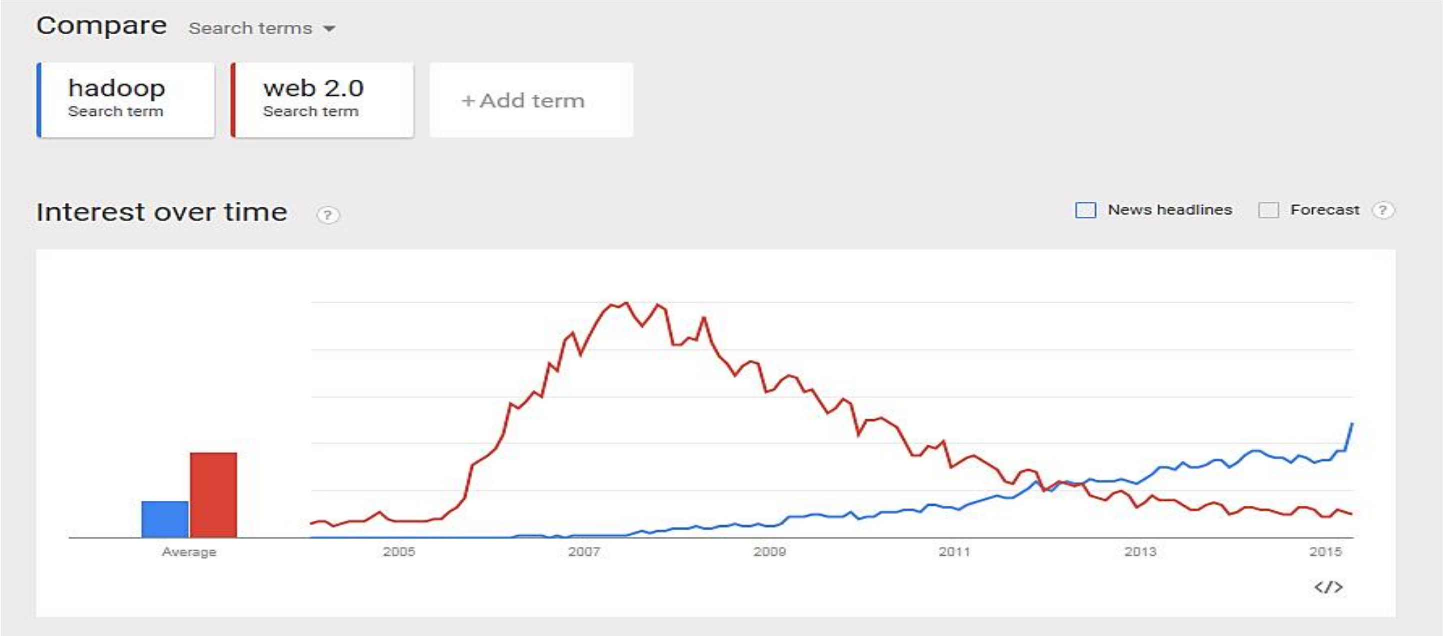


Figure 9-24 Google Trends on the Terms Web 2.0 and Hadoop
Source: Google and the Google logo are registered trademarks of Google Inc., Used with permission.

Hadoop

Q9-6 How do organizations use Big Data applications?

- Open-source program supported by Apache Foundation2.
- Manages thousands of computers.
- Implements MapReduce.
  - Written in Java.
- Amazon.com supports Hadoop as part of EC3 cloud.
- Query language entitled Pig (platform for large dataset analysis).
  - Easy to master.
  - Extensible.
  - Automatically optimizes queries on map-reduce level.

Knowledge Management Systems

Q9-7 What is the role of knowledge management systems?

- Knowledge Management (KM)
  - Creating value from intellectual capital and sharing knowledge with those who need that capital.
- Preserving organizational memory
  - Capturing and storing lessons learned and best practices of key employees.
- Scope of KM same as SM in hyper-social organizations.

Benefits of Knowledge Management

Q9-7 What is the role of knowledge management systems?

- Improve process quality.
- Increase team strength.
- Goal:
  - Enable employees to use organization’s collective knowledge.

What Are Expert Systems?

Q9-7 What is the role of knowledge management systems?

- Expert systems
  - Rule-based IF/THEN
  - Encode human knowledge
- Expert systems shells
  - Process IF side of rules
  - Report values of all variables
  - Knowledge gathered from human experts

Example of IF/THEN Rules

Q9-7 What is the role of knowledge management systems?

```
Other rules here...

IF CardiacRiskFactor = 'Null' THEN Set CardiacRiskFactor = 0
IF PatientSex = 'Male' THEN Add 3 to CardiacRiskFactor
IF PatientAge >55 THEN Add 2 to CardiacRiskFactor
IF FamilyHeartHistory = 'True' THEN Add 5 to CardiacRiskFactor
IF CholesterolScore = 'Problematic' THEN Add 4 to CardiacRiskFactor
IF BloodPressure = 'Problematic' THEN Add 3 to CardiacRiskFactor
IF CardiacRiskFactor >15 THEN Set EchoCardiagramTest = 'Schedule'
...
Other rules here...
```

Figure 9-25 Example of If/Then Rules

Drawbacks of Expert Systems

1. Difficult and expensive to develop.
   – Labor intensive.
   – Ties up domain experts.
2. Difficult to maintain.
   – Changes cause unpredictable outcomes.
   – Constantly need expensive changes.
3. Don’t live up to expectations.
   – Can’t duplicate diagnostic abilities of humans.

What Are Content Management Systems (CMS)?

- Support management and delivery of documents, other expressions of employee knowledge.
- Challenges of Content Management
  - Huge databases.
  - Dynamic content.
  - Documents refer to one another.
  - Perishable contents.
  - In many languages.

What are CMS Application Alternatives?

- In-house custom development
  - Customer support develops in-house database applications to track customer problems.
- Off-the-shelf
  - Horizontal market products (SharePoint).
  - Vertical market applications.
- Public search engine
  - Google, Bing.

How Do Hyper-Social Organizations Manage Knowledge?

- **Hyper-social knowledge management**
  - Social media, and related applications, for management and delivery of organizational knowledge resources.
- **Hyper-organization theory**
  - Framework for understanding KM.
  - Focus shifts from knowledge and content to fostering authentic relationships among knowledge creators and users.

Hyper-Social KM Media

Media	Public or Private	Best for:
Blogs	Either	Defender of belief
Discussion groups (including FAQ)	Either	Problem solving
Wikis	Either	Either
Surveys	Either	Problem solving
Rich directories (e.g. Active Directory)	Private	Problem solving
Standard SM (Facebook, Twitter, etc.)	Public	Defender of belief
YouTube	Public	Either

Figure 9-27 Hyper-Social KM Media

Resistance to Knowledge Sharing

- Employees reluctant to exhibit their ignorance.
- Employee competition.
- Remedy
  - Strong management endorsement.
  - Strong positive feedback.
  - “Nothing wrong with praise or cash . . . especially cash.”

BI Publishing Alternatives

Q9-8 What are the alternatives for publishing BI?

Server	Report Type	Push Options	Skill Level Needed
Email or collaboration tool	Static	Manual	Low
Web server	Static/Dynamic	Alert/RSS	Low for static High for dynamic
SharePoint	Static/Dynamic	Alert/RSS Workflow	Low for static High for dynamic
BI server	Dynamic	Alert/RSS Subscription	High

Figure 9-28 BI Publishing Alternatives

What Are the Two Functions of a BI Server?

Q9-8 What are the alternatives for publishing BI?

- Management and delivery

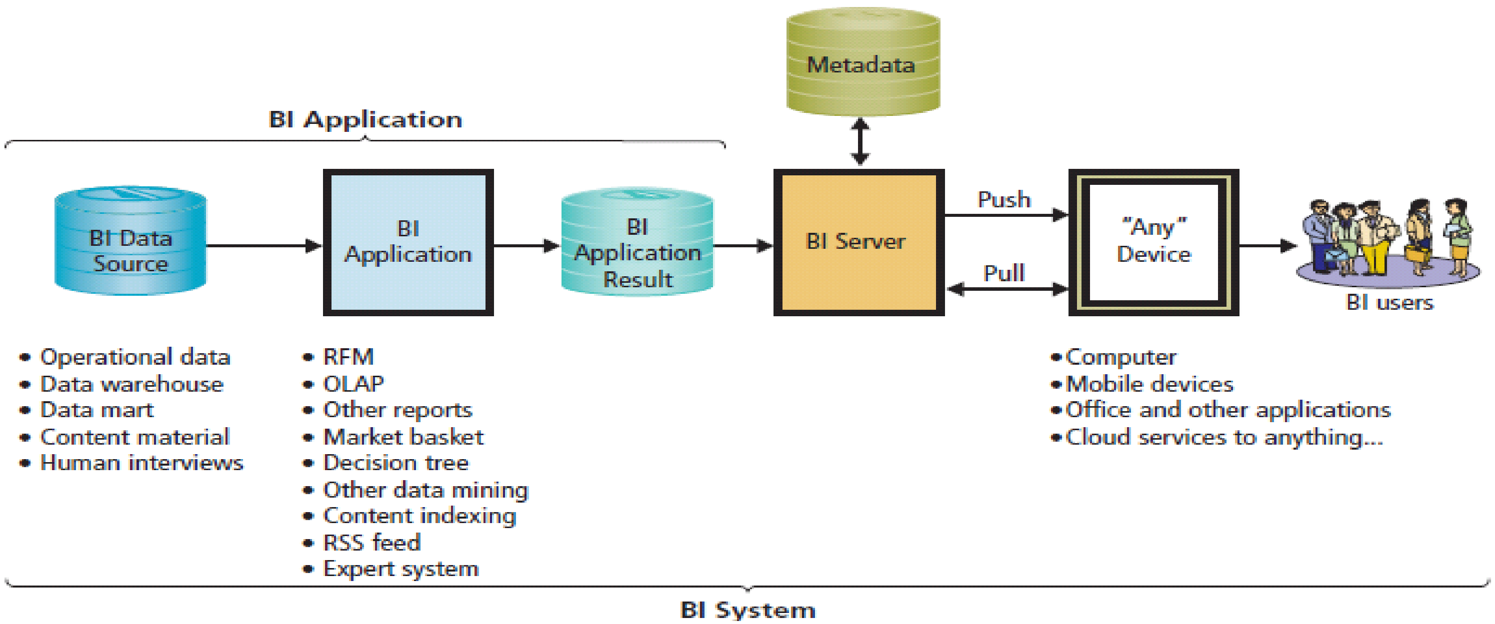


Figure 9-29 Elements of a BI System

Business Intelligence Systems in 2027

Q9-9 2027?

- Exponentially more information about customers, better data mining techniques.
- Companies buy and sell your purchasing habits and psyche.
- **Singularity**
  - Computer systems adapt and create their own software without human assistance.
  - Machines will possess and create information for themselves.
  - Will we know what the machines will know?

Semantic Security

Security Guide

1. Unauthorized access to protected data and information.
   - Physical security
     - Passwords and permissions.
     - Delivery system must be secure.
2. Unintended release of protected information through reports and documents.
3. What, if anything, can be done to prevent what Megan did?

Manager, Data and Analytics

Career Guide

Lindsey Tsuya at American Express Company

Q. What attracted you to this field?

A. “As a college student, I worked in the service industry. When I was selecting my degree, I knew I wanted two things. First, I wanted a degree that made money. Second, I wanted a job that did not involve direct provision of service to the public. By choosing information systems, I knew I would be doing more of a behind-the-scenes job.”

Q. What advice would you give to someone who is considering working in your field?

A. “No matter what field you choose, make sure it is something you are passionate about because if you are not passionate about it, work will feel like… work.”

Active Review

Q9-1 How do organizations use business intelligence (BI) systems?

Q9-2 What are the three primary activities in the BI process?

Q9-3 How do organizations use data warehouses and data marts to acquire data?

Q9-4 How do organizations use reporting applications?

Q9-5 How do organizations use data mining applications?

Q9-6 How do organizations use Big Data applications?

Q9-7 What is the role of knowledge management systems?

Q9-8 What are the alternatives for publishing BI?

Q9-9 2027?

Hadoop the Cookie Cutter

Case Study 9

- Third-party cookie created by site other than one you visited.
- Most commonly occurs when a Web page includes content from multiple sources.
- DoubleClick
  - IP address where content was delivered.
    - DoubleClick instructs your browser to store a DoubleClick cookie.
  - Records data in cookie log on DoubleClick's server.

Hadoop the Cookie Cutter (cont'd)

Case Study 9

- Third-party cookie owner has history of what was shown, what ads you clicked, and intervals between interactions.
- Cookie log shows how you respond to ads and your pattern of visiting various Web sites where ads placed.
- Firefox Lightbeam tracks and graphs cookies on your computer.

FireFox Lightbeam: Display on Start Up

Case Study 9

- No cookies on startup.

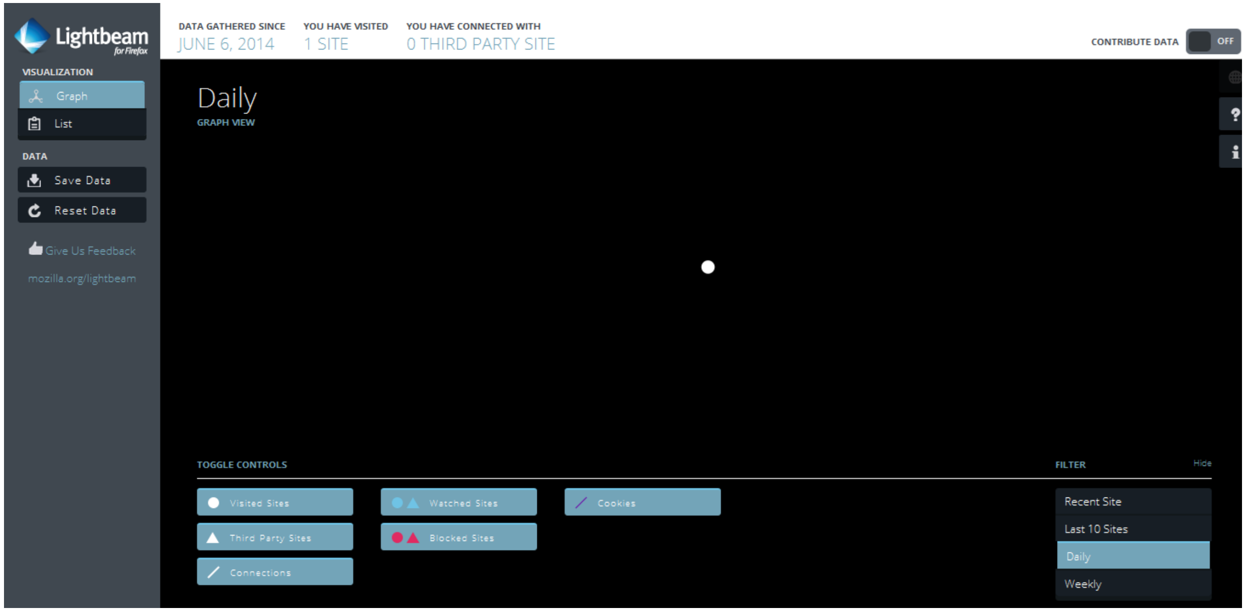


Figure 9-30a Third-Party Cookie Growth
Source: © Mozilla Corporation

After Visiting MSN.com

Case Study 9

- After MSN.com and Gmail.

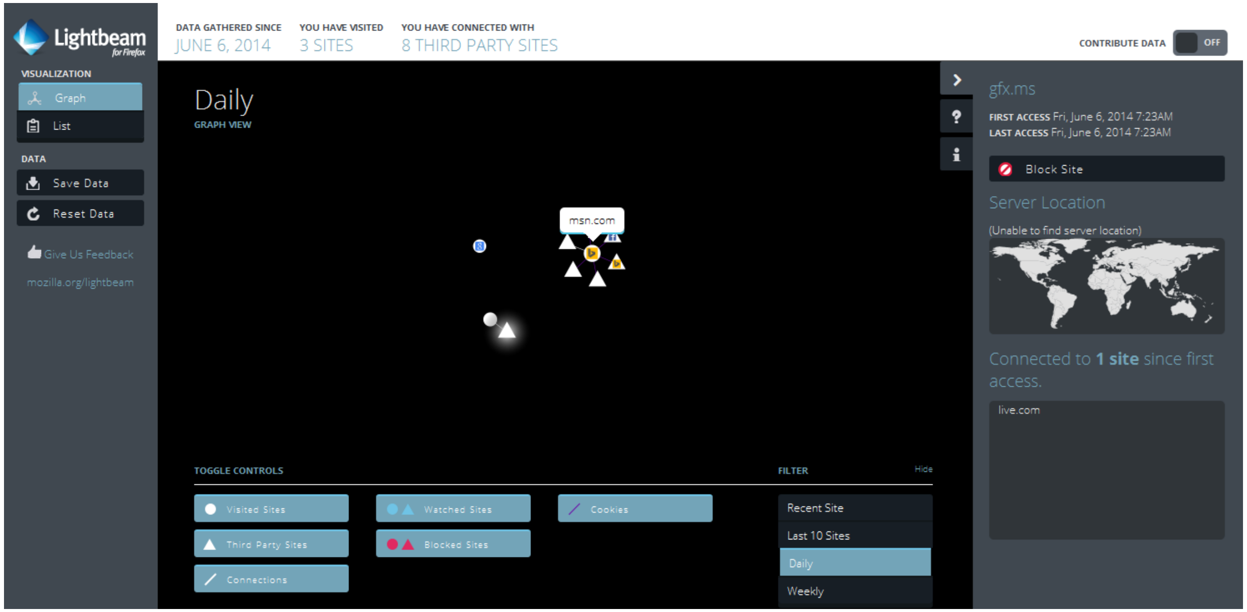


Figure 9-30b Third-Party Cookie Growth
Source: © Mozilla Corporation

5 Sites Visited Yields 27 Third Parties

Case Study 9

- Five sites visited yield 27 third parties.

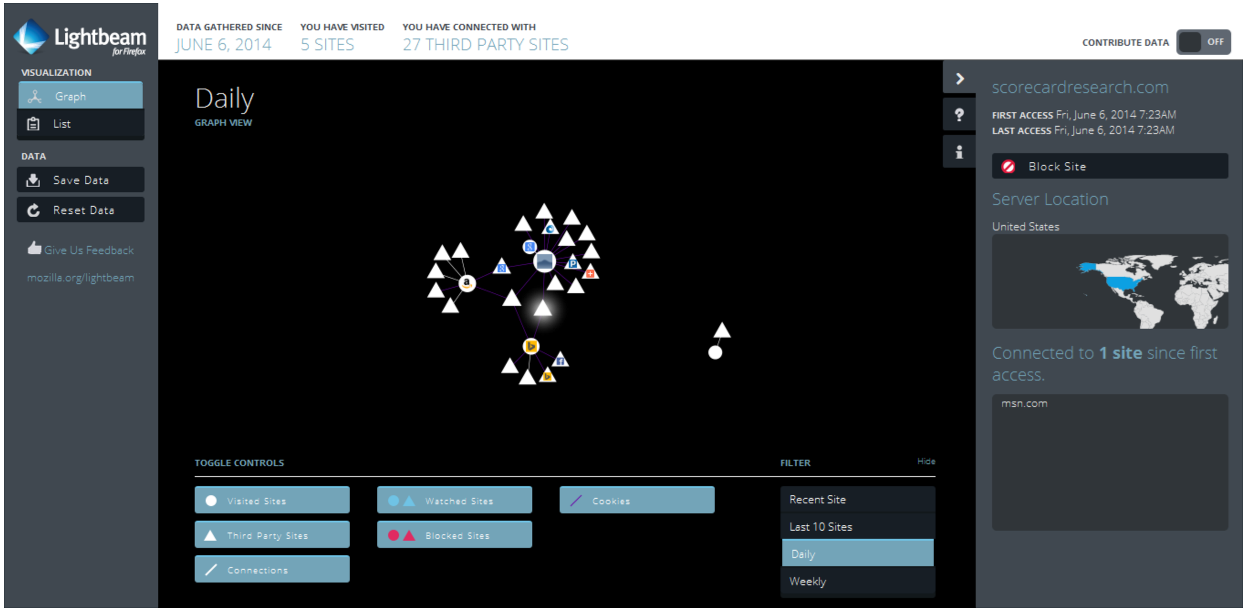


Figure 9-30c Third-Party Cookie Growth
Source: © Mozilla Corporation

Sites Connected to DoubleClick

Case Study 9

- Sites connected to DoubleClick.

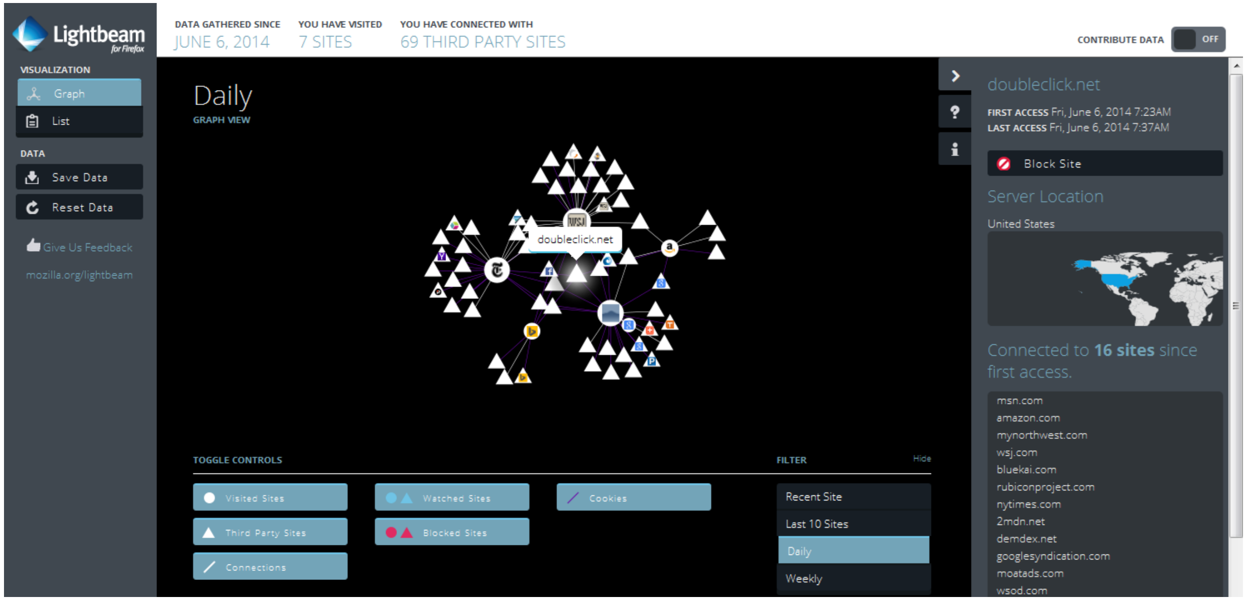


Figure 9-30d Third-Party Cookie Growth
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